

Q.10.1

Data Set 1 $\bar{y} = 7.017$

X	Y	X ²	X ³	X ⁴	XY	X ² Y	$\hat{Y}_{reg}$	$\hat{Y}_{poly}$	
3	2.5	9	27	81	7.5	22.5	0.723	2.490	3
4	3.2	16	64	256	12.8	51.2	3.240	2.896	4
5	3.8	25	125	625	19.0	95.0	5.757	4.359	5
6	6.5	36	216	1296	39.0	234.0	8.274	6.882	6
7	11.5	49	343	2401	80.5	563.5	10.791	10.463	7
8	14.6	64	512	4096	116.8	934.4		15.102	8
33	42.1	199	1287	8755	275.1	1901.6			

Linear Regression

$$y = b_0 + b_1x$$

$$b_1 = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2} = \frac{6(275.1) - (33)(42.1)}{6(199) - (33)^2}$$

$$b_1 = 2.52$$

$$b_0 = \frac{\sum Y - b_1 \sum X}{n} = \frac{42.1 - (2.52)(33)}{6}$$

$$b_0 = -6.883$$

$$\hat{y}_{reg} = -6.883 + 2.52x$$

R^2

SSR

SST

$$SSR = \sum (\hat{y}_{reg} - \bar{y})^2$$

$$= 39.614 + 14.266 + 15.88 + 15.80 + 14.243 + 39.577$$

$$SST = \sum (y - \bar{y})^2$$

$$= 20.403 + 14.569 + 10.349 = 110.868$$

$$+ 0.267 + 20.097 + 57.502$$

$$= 123.188$$

$$R^2 = \frac{110.868}{123.188} \Rightarrow 0.9001$$

Polynomial Regression (deg 2): $y = b_0 + b_1x + b_2x^2$

$$\sum Y = nb_0 + (\sum X)b_1 + (\sum X^2)b_2 \Rightarrow 42.1 = 6b_0 + 33b_1 + 199b_2$$

$$\sum YX = (\sum X)b_0 + (\sum X^2)b_1 + (\sum X^3)b_2 \Rightarrow 275.1 = 33b_0 + 199b_1 + 1287b_2$$

$$\sum YX^2 = (\sum X^2)b_0 + (\sum X^3)b_1 + (\sum X^4)b_2 \Rightarrow 1901.6 = 199b_0 + 1287b_1 + 8755b_2$$

After calculation

$$b_0 = 7.62 \quad b_1 = -3.30 \quad b_2 = 0.53$$

$$\hat{y}_{poly} = 7.62 - 3.30X + 0.53X^2$$

R^2 %

$$\frac{SSR_{poly}}{SST}$$

$$SST = 123.188$$

$$\begin{aligned} SSR &= 20.494 + 16.983 + 7.065 + 0.018 + 11.875 + 65.367 \\ &= 121.802 \end{aligned}$$

$$R^2 = 0.9848$$

This shows that polynomial regression explain about 98.48% of Error in Dataset 1 and ~~provide an reasonable fit~~ capture an increasing trend of X in dataset 1 then linear regression ($R^2 = 0.9001$).

Q110.1

Dataset 2:

$$\bar{Y} = 7.75$$

X	Y	X ²	X ³	X ⁴	XY	X ² Y	$\hat{Y}_{reg}$	$\hat{Y}_{poly}$
1	2	1	1	1	2	2	0.7	1.95
2	4	4	8	16	8	16	5.4	4.15
3	9	9	27	81	27	81	10.1	8.85
4	16	16	64	256	64	256	14.8	16.05
10	31	30	100	354	101	355		

Linear Regression: $\hat{y} = b_0 + b_1x$

$$b_1 = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

$$b_0 = \frac{\sum Y - b_1 \sum X}{n}$$

$$= \frac{4(101) - (10)(31)}{4(30) - (10)^2} = \frac{31 - (4.70)(10)}{4}$$

$$= 4.70 \quad = -4.00$$

$$\hat{y} = -4.00 + 4.70X$$

 R^2 :

$$= \frac{SSR}{SST} \Rightarrow \frac{(-7.05)^2 + (-2.35)^2 + (2.35)^2 + (7.05)^2}{(2-7.75)^2 + (4-7.75)^2 + (9-7.75)^2 + (16-7.75)^2}$$

$$SSR = \sum (\hat{y}_{reg} - \bar{y})^2 \Rightarrow 0.9460$$

Polynomial Regression (deg 2):

$$\hat{y}_{poly} = b_0 + b_1x + b_2x^2$$

$$\sum Y = nb_0 + b_1 \sum X + b_2 \sum X^2$$

$$\sum YX = \sum X(b_0) + b_1 \sum X^2 + b_2 \sum X^3$$

$$\sum YX^2 = \sum X^2(b_0) + b_1 \sum X^3 + \sum X^4(b_2)$$

$$31 = 4b_0 + 10b_1 + 30b_2$$

$$101 = 10b_0 + 30b_1 + 100b_2$$

$$355 = 30b_0 + 100b_1 + 354b_2$$

After solving

$$b_0 = 2.25 \quad b_1 = -1.55 \quad b_2 = 1.25$$

$$\hat{y}_{poly} = 2.25 - 1.55X + 1.25X^2$$

R^2 :

$$\frac{SSR_{poly}}{SST} = \frac{(-5.8)^2 + (-3.6)^2 + (1.1)^2 + (8.3)^2}{116.75}$$
$$= 0.9996$$

This shows that polynomial Regression explains about 99.96% error while LR explains 94.60%. Hence polynomial regression outperform the LR and capture the appropriate trend.

Q NO. 4

Aptitude

Algorithm

Item set 1

Sup min = 3

① A1 2 ②

① A5 8 ②

A2 6

A7 7

A3 6

A2 6

A4 4

A3 6

A5 8 ✓

A6 5

A6 5

A4 4

A7 7 ✓

A8 4

A8 4

A9 2

Item set 2

① A5 A7 5 ②

① A7 A2 2 ②

① A2 A6 2 ②

① A6 A4 1 ②

~~A5 A2 0~~

A7 A3 3

A2 A4 3

~~A6 A8 2~~

A5 A3 3

~~A7 A6 3~~~~A2 A8 1~~~~A4 A8 1~~

A5 A6 3

~~A7 A4 2~~~~A3 A6 0~~~~A5 A4 1~~~~A7 A8 0~~~~A3 A4 1~~~~A5 A8 2~~

A2 A3 3

~~A3 A8 0~~

① A5 A7 5 ②

Item set 3

✓ A5 A3 3

① A5 A7 A3 3 ②

① A2 A3 A7 0 ②

✓ A5 A6 3

~~A5 A7 A6 1~~~~A2 A3 A4 A7 0~~

✓ A7 A3 3

A2 A3 A5 A7 0

~~A3 A2 A6 A7 0~~

✓ A7 A6 3

A2 A4 A5 A7 0

~~A2 A4 A6 A7 1~~

A2 A3 3

A5 A3 A6 0

~~A2 A3 A4 1~~

A2 A4 3

A2 A3 A5 0

A2 A3 A4 A5 0

A3 A5 A6 A7 0

A2 A3 A5 A6 0

A2 A4 A5 A6 0

A3 A6 A7 0

Item set 4

① A5 A7 A3 3 ②

Rules :

$$A3 \ A5 \rightarrow A7$$

$$A5 \ A7 \rightarrow A3$$

$$A3 \ A7 \rightarrow A5$$

Confidence

$$3/3 = 1$$

$$3/5 = 0.6$$

$$3/3 = 1$$

Lift :-

$A3 \ A5 \rightarrow A7$

Probabilities

$$L = \frac{0.2}{0.2 \times 0.467}$$

$$= 2.141$$

$$A3 \ A5 \ A7 = 3/15 \Rightarrow 0.2$$

$$A3 \ A5 = 3/15 \Rightarrow 0.2$$

$$A7 = 7/15 \Rightarrow 0.467$$

$A5 \ A7 \rightarrow A3$

$$L = \frac{0.2}{0.33 \times 0.467}$$

$$= 1.317$$

$$A3 \ A5 \ A7 = 0.2$$

$$A5 \ A7 = 0.33$$

$$A3 = 0.467$$

$A3 \ A7 \rightarrow A5$

$$L = \frac{0.2}{0.2 \times 0.53}$$

$$= 1.88$$

$$A3 \ A5 \ A7 = 0.2$$

$$A3 \ A7 = 0.2$$

$$A5 = 0.53$$

Conclusion :

The confidence shows that there is 0.6 (60%) chance that A3 item will be bought when A5, A7 is bought and so on for other rules. All ~~positive~~ values of lift is greater than 1 shows positive association

Arrange items in order

A5 A7 A3 A2 A6 A4 A8

Tree %

